

Design and Performance Evaluation of Free Space Optics and Fiber optic Communication System

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*A Project Report submitted in partial fulfillment of the requirements for award of
the degree*

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CERTIFICATE

This is to certify that the project report entitled

**“Design and Performance
Evaluation of Free Space Optics
and Fiber optic Communication
System”**

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Abstract

This project focuses on the comparative analysis of Free Space Optics (FSO) and Optical Fiber communication systems using OptiSystem simulation software. Both technologies are widely used for high-speed data transmission, but their performance varies under different operating conditions. The main objective of this project is to evaluate and compare the performance of these two communication systems by varying parameters such as transmission distance and attenuation.

The system is designed using components such as PRBS generator, NRZ pulse generator, Mach-Zehnder modulator with continuous wave laser source, optical amplifier, transmission channel (FSO or Optical Fiber), EDFA, PIN photodetector, low pass filter, and 3R regenerator. The simulation is carried out at a bit rate of 10 Gbps and a wavelength of 193.1 THz. For the FSO system, different attenuation values such as 3 dB/km, 5 dB/km, 10 dB/km, and 15 dB/km are considered to analyze system performance under varying conditions.

The performance of both systems is evaluated using parameters like Q-factor, Bit Error Rate (BER), and eye diagram. From the simulation results, it is observed that Optical Fiber communication provides more reliable and stable performance compared to FSO, especially at longer distances and higher attenuation values.

The project highlights the limitations of FSO communication under adverse conditions and demonstrates the effectiveness of Optical Fiber communication for long-distance data transmission. The use of simulation tools helps in analyzing system performance accurately and provides a better understanding of both technologies.

Keywords: FSO, Optical Fiber, OptiSystem, Q-factor, BER, Eye Diagram, Attenuation, Communication System

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Chapter 1 Introduction

1.1 Introduction

In modern communication systems, high-speed and reliable data transmission is very important. Technologies like Free Space Optics (FSO) and Optical Fiber communication are widely used to transmit large amounts of data over long distances. However, the performance of these systems varies depending on environmental conditions, transmission distance, and signal attenuation. Therefore, it is necessary to analyze and compare these technologies to understand their advantages and limitations.

This project focuses on the comparative analysis of FSO and Optical Fiber communication systems using OptiSystem simulation software. The system is designed using components such as PRBS generator, NRZ pulse generator, Mach-Zehnder modulator, optical amplifier, transmission channel, and receiver section. Both communication systems are tested under different conditions by varying parameters like distance and attenuation.

To evaluate the system performance, parameters such as Q-factor, Bit Error Rate (BER), and eye diagram are analyzed. In the case of FSO communication, attenuation plays a significant role in affecting performance, while Optical Fiber communication provides more stable transmission under similar conditions.

By using simulation tools and analyzing different parameters, this project helps in understanding the behavior of both communication systems. It provides a clear comparison between FSO and Optical Fiber, which is useful for selecting the appropriate technology based on application requirements.

1.2 Aim

To analyze and compare the performance of Free Space Optics (FSO) and Optical Fiber communication systems using OptiSystem software by evaluating parameters such as Q-factor, Bit Error Rate (BER), and eye diagram under varying conditions of transmission distance and attenuation.

1.3 Objectives

To design and simulate Free Space Optics (FSO) and Optical Fiber communication systems using OptiSystem software, and to analyze their performance under different operating conditions. The project aims to evaluate and compare both communication technologies based on important performance parameters, while addressing the limitations of FSO communication such as high attenuation sensitivity and limited transmission range. Following are some goals of the project:

- To analyze system performance at a bit rate of 10 Gbps and wavelength of 193.1 THz
- To study the effect of transmission distance on both FSO and Optical Fiber systems
- To evaluate the impact of attenuation in FSO and Optical Fiber system (3 dB/km, 5 dB/km, 10 dB/km, 15 dB/km)
- To measure parameters such as Q-factor, Bit Error Rate (BER), eye height, and threshold
- To identify the more reliable communication system for long-distance transmission

1.1 Block Diagram

1.4.1 Free Space Optical Communication Channel

The block diagram of FSO consists of an optical transmitter, free-space atmospheric channel, and optical receiver. The optical signal travels wirelessly through air using a laser beam.

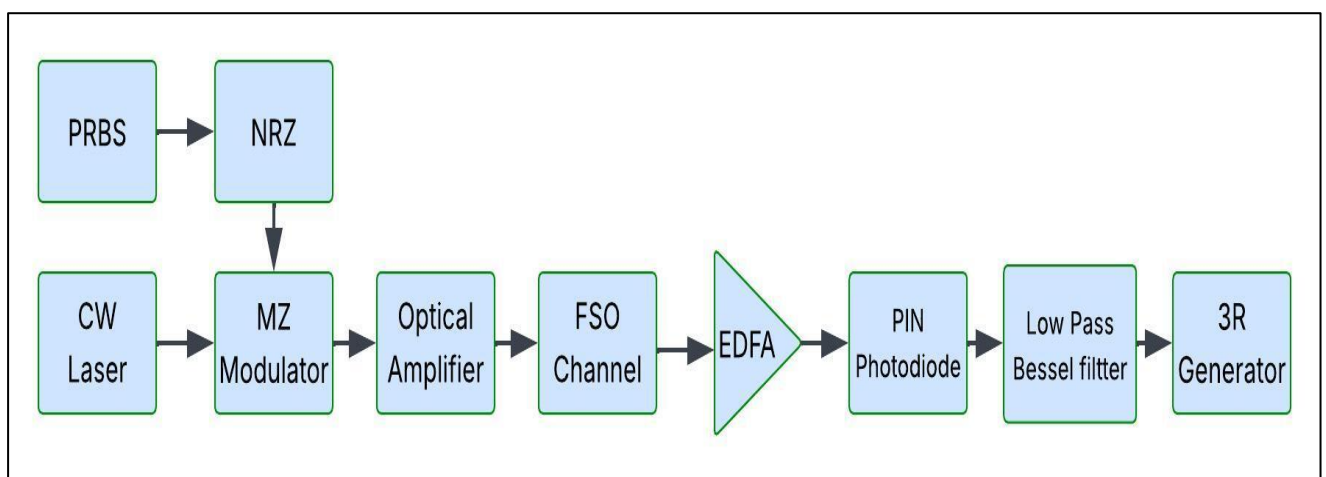


Figure 1.1 : Block Diagram of FSO Channel

1.4.2 Optical Fiber Communication Channel

The block diagram of OFC consists of an optical transmitter, optical fiber cable, and photodetector receiver. The optical signal travels through the fiber using total internal reflection.

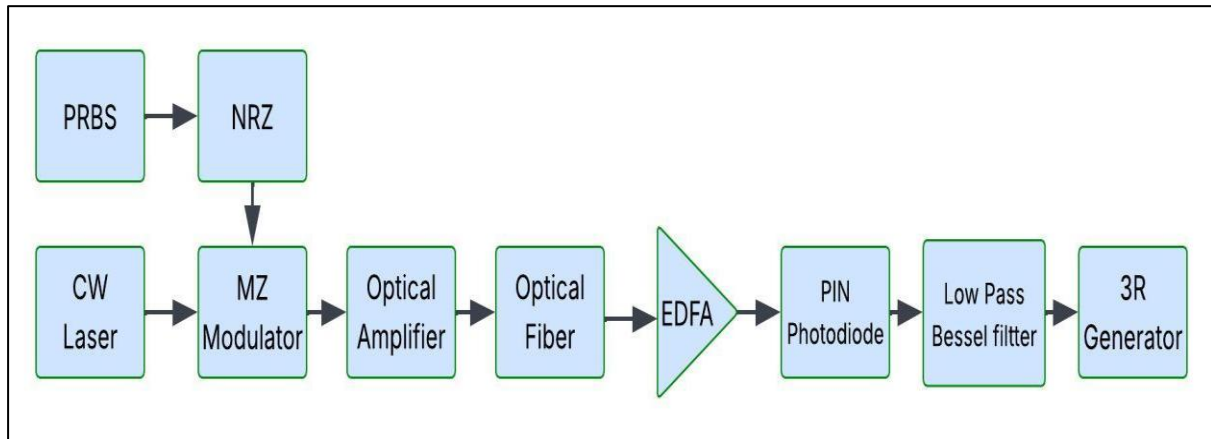


Figure 1.2 : Block Diagram of Optical Fiber Channel

Feature	FSO (Free Space Optics)	Optical Fiber Communication
Transmission Medium	Air / free space	Optical fiber cable
Signal Transmission	Wireless optical beam	Guided through fiber
Main Difference	free-space atmospheric channel	optical fiber link
Weather Effect	Affected by fog, rain, dust	Less affected by weather
Installation	Easy, no cable required	Requires fiber cable installation

Table 1.1 Comparison table of FSO and Optical Fiber

Explanation of Block Diagram:

PRBS (Pseudo-Random Bit Sequence):

The Pseudo-Random Bit Sequence (PRBS) generator is the initial block of the optical communication system, responsible for producing a binary data stream that closely resembles real-world digital communication signals. Instead of using a fixed or repetitive pattern, PRBS generates a sequence of bits (0s and 1s) that appear random but are actually deterministic and repeat after a certain length. The purpose of using PRBS is to simulate realistic data traffic

conditions, enabling accurate testing and evaluation of system performance under varying transmission scenarios.

NRZ (Non-Return-to-Zero):

The binary data generated from the PRBS is then fed to the NRZ encoder, which converts it into an electrical signal suitable for transmission. In this method, a high level represents a binary '1' and a low level represents a binary '0', with the signal remaining constant throughout the bit duration. This encoding technique ensures efficient bandwidth utilization and provides a simple way to prepare the signal for optical modulation in OptiSystem.

CW Laser (Continuous Wave Laser):

The encoded electrical signal is used to control a Continuous Wave (CW) laser, which acts as the optical source in the system. The CW laser generates a constant optical signal with a fixed wavelength and power level, providing a stable carrier for data transmission. This continuous light output is essential for efficient modulation and reliable performance in optical communication systems simulated using OptiSystem.

MZ Modulator (Mach-Zehnder Modulator):

The electrical NRZ signal is then applied to the Mach-Zehnder (MZ) modulator, which is used to modulate the optical signal generated by the CW laser. It works by varying the intensity of the light based on the input electrical signal, effectively converting the electrical data into an optical form. This modulation technique provides high-speed operation and is widely used for efficient and accurate signal transmission in systems designed using OptiSystem.

Optical Amplifier:

The modulated optical signal is then passed through an optical amplifier, which increases the signal power before transmission. This amplification helps to compensate for losses and ensures that the signal can travel longer distances without significant degradation. It improves overall system performance and maintains signal quality in optical communication systems simulated using OptiSystem.

FSO (Free Space Optics):

The amplified optical signal is then transmitted through the Free Space Optics (FSO) channel, where the signal propagates through the atmosphere without the use of physical fiber. This wireless optical communication method uses line-of-sight transmission and is affected by environmental conditions such as fog, rain, and atmospheric turbulence. FSO enables high data rate communication over short to medium distances and is analyzed in OptiSystem by varying parameters like range and attenuation.

Optical Fiber Cable:

The amplified optical signal can alternatively be transmitted through an optical fiber cable, which guides light through a core using total internal reflection. This method provides low-loss, high-speed, and long-distance communication with minimal interference. Optical fiber ensures reliable signal transmission and is analyzed in OptiSystem to compare performance parameters like Q-factor, BER, and eye diagram against FSO.

EDFA (Erbium-Doped Fiber Amplifier):

The optical signal received from the FSO or fiber channel is then passed through an EDFA, which amplifies the light without converting it to an electrical signal. The EDFA boosts the signal power to compensate for transmission losses, maintaining signal strength and quality over long distances. It is widely used in optical communication systems simulated in OptiSystem for efficient long-range data transmission.

PIN Photodiode:

The amplified optical signal is then received by a PIN photodiode, which converts the incoming light back into an electrical signal. The PIN structure provides fast response and high sensitivity, making it ideal for detecting modulated optical signals. In OptiSystem, the PIN photodiode is used to recover the transmitted data accurately for further processing and analysis.

Low-Pass Filter:

The electrical signal from the PIN photodiode is fed into a low-pass filter, which eliminates high-frequency noise and unwanted distortions. This process smooths the signal and preserves the integrity of the transmitted data, ensuring accurate detection. In OptiSystem, the low-pass filter enhances the quality of the signal for reliable evaluation of Q-factor, BER, and eye

diagram.

3R Generator (Re-amplification, Re-shaping, Re-timing):

The filtered electrical signal is then processed by a 3R generator, which performs Re-amplification, Re-shaping, and Re-timing of the data. This restores the signal's amplitude, corrects waveform distortions, and aligns the timing of each bit, ensuring accurate data recovery. In OptiSystem, the 3R generator is crucial for maintaining signal integrity over long-distance FSO or fiber communication links.

Chapter 2 Literature Review

2.1 Introduction

The literature review provides an overview of existing research on Free Space Optics (FSO) and Optical Fiber communication systems, which are widely used for high-speed data transmission. Previous studies show that Optical Fiber offers stable and low-loss communication over long distances, while FSO provides a flexible and cost-effective wireless alternative but is highly affected by atmospheric conditions like fog and rain, leading to signal attenuation. Researchers have commonly used simulation tools such as OptiSystem to analyze system performance using parameters like Q-factor, Bit Error Rate (BER), and eye diagram. These studies help in understanding the strengths and limitations of both technologies and form the basis for the comparative analysis carried out in this project.

2.2 Literature Review

1. Title: “Free Space Optics: Enabling Optical Connectivity in Today’s Networks”

- **Authors:** H. Willebrand, B. S. Ghuman
- **Published by:** Sams Publishing, 2002
- **Summary:** This book explains the fundamentals and applications of Free Space Optical (FSO) communication systems. The authors discussed the advantages of FSO technology such as high bandwidth, fast deployment, low installation cost, and license-free operation. The book also highlights challenges like atmospheric attenuation, alignment issues, and weather effects on signal transmission. In my project, these concepts are used to study the effect of attenuation on FSO system performance using OptiSystem software and compare it with optical fiber communication.

2. Title: “Free-Space Laser Communications: Principles and Advances”

- **Authors:** A. Majumdar, J. C. Ricklin
- **Published by:** Springer, 2008
- **Summary:** This book provides detailed information about free-space laser communication systems, including propagation theory, atmospheric effects, modulation techniques, and system design. It explains how environmental conditions influence signal quality and communication reliability. The authors also discussed performance analysis using parameters such as BER and signal-to-noise ratio. In my

project, these principles are applied to analyze the performance degradation of FSO systems under different attenuation conditions.

3. Title: “Fiber-Optic Communication Systems”

- **Authors:** G. P. Agrawal
- **Published by:** Wiley, 2010
- **Summary:** This book explains the principles of fiber-optic communication systems, including optical transmission, attenuation, dispersion, modulation techniques, and receiver design. It also discusses the advantages of optical fiber such as low signal loss, high bandwidth, and stable long-distance communication. In my project, these concepts are used to evaluate and compare the stable performance of optical fiber communication systems with FSO systems under varying transmission distances.

4. Title: “Optical Fiber Communications: Principles and Practice”

- **Authors:** J. M. Senior, M. Y. Jamro
- **Published by:** Pearson, 2009
- **Summary:** The book describes the working principles of optical fiber communication systems and explains important parameters such as attenuation, dispersion, bandwidth, and optical receiver performance. It also provides knowledge about optical system components and signal transmission techniques. In my project, the concepts discussed in this book help in designing and simulating the optical fiber communication model using OptiSystem software.

5. Title: “Free-space optical communication through atmospheric turbulence channels”

- **Authors:** X. Zhu, J. M. Kahn
- **Published by:** IEEE Transactions on Communications, 2002
- **Summary:** This paper analyzes the effect of atmospheric turbulence on Free Space Optical communication systems. The authors explained how turbulence causes fluctuations in optical signal intensity, leading to increased BER and reduced communication reliability. The study also discusses methods to improve system performance under atmospheric disturbances. In my project, similar atmospheric attenuation effects are considered while analyzing FSO system performance at different attenuation levels.

6. Title: “Optical communication in space: Challenges and mitigation techniques”

- **Authors:** H. Kaushal, G. Kaddoum
- **Published by:** IEEE Communications Surveys & Tutorials, 2017
- **Summary:** This paper discusses optical communication systems used in space applications and explains various challenges such as atmospheric loss, pointing errors, turbulence, and signal fading. The authors also presented different mitigation techniques to improve communication reliability and performance. In my project, this study supports the understanding of how attenuation and environmental conditions affect FSO communication systems.

7. Title: “Comparison of laser beam propagation in fog and haze for optical wireless communications”

- **Authors:** I. I. Kim, B. McArthur, E. J. Korevaar
- **Published by:** SPIE Proceedings, 2001
- **Summary:** This research compares the propagation characteristics of laser beams in different weather conditions such as fog and haze. The study showed that fog causes severe attenuation in FSO systems, significantly reducing communication range and signal quality. In my project, similar attenuation conditions are simulated to study their effect on BER and Q-factor in FSO communication systems.

8. Title: “OptiSystem User Reference”

- **Authors:** Optiwave Systems Inc.
- **Published by:** Optiwave Systems Inc.
- **Summary:** This reference manual explains the features, components, and simulation techniques available in OptiSystem software. It provides guidelines for designing optical communication systems using components such as PRBS generators, NRZ pulse generators, Mach–Zehnder modulators, optical amplifiers, and BER analyzers. In my project, OptiSystem software is used to simulate and analyze both FSO and optical fiber communication systems.

9. Title: “Characteristics of a Single-Mode Optical Fibre and Cable”

- **Authors:** ITU-T Recommendation G.652
- **Published by:** International Telecommunication Union (ITU), 2016
- **Summary:** This recommendation defines the standard characteristics of single-mode

optical fibers used in communication systems. It includes specifications related to attenuation, dispersion, operating wavelength, and transmission performance. The standard ensures reliable and efficient optical communication over long distances. In my project, these standard optical fiber characteristics are considered while designing the optical fiber communication system in OptiSystem.

10. Title: “Optical Wireless Communication Systems”

- **Authors:** S. Arnon
- **Published by:** Optical Engineering, 2003
- **Summary:** This paper discusses the principles, applications, and challenges of optical wireless communication systems, especially Free Space Optics. It explains system design, atmospheric effects, modulation techniques, and performance analysis methods. The study highlights the importance of BER and signal quality evaluation under different environmental conditions. In my project, these concepts are applied to analyze and compare the performance of FSO and optical fiber communication systems using simulation techniques.

2.3 Literature Review Summary

Title	Authors	Publisher	Summary
Free Space Optics: Enabling Optical Connectivity in Today's Networks	H. Willebrand, B. S. Ghuman	Sams Publishing	Explains fundamentals, advantages, and challenges of FSO systems, including atmospheric attenuation and signal degradation effects.
Free-Space Laser Communications: Principles and Advances	A. Majumdar, J. C. Ricklin	Springer	Discusses principles of free-space laser communication, atmospheric effects, and performance evaluation using BER and signal quality analysis.
Fiber-Optic Communication Systems	G. P. Agrawal	Wiley	Describes optical fiber communication principles, attenuation, dispersion, and advantages such as low loss and stable long-distance transmission.
Optical Fiber Communications: Principles and Practice	J. M. Senior, M. Y. Jamro	Pearson	Explains optical fiber communication components, transmission characteristics, attenuation, and receiver performance.
Free-space optical communication through atmospheric turbulence channels	X. Zhu, J. M. Kahn	IEEE Transactions on Communications	Analyzes atmospheric turbulence effects on FSO systems and shows its impact on BER and communication reliability.
Optical communication in space: Challenges and mitigation techniques	H. Kaushal, G. Kaddoum	IEEE Communications Surveys & Tutorials	Discusses challenges in optical space communication such as turbulence, attenuation, and pointing errors along

			with mitigation techniques.
Comparison of laser beam propagation in fog and haze for optical wireless communications	I. I. Kim, B. McArthur, E. J. Korevaar	SPIE Proceedings	Compares laser propagation in fog and haze conditions and explains severe attenuation effects in FSO communication systems.
OptiSystem User Reference	Optiwave Systems Inc.	Optiwave Systems Inc.	Provides guidelines for designing and simulating optical communication systems using OptiSystem components and analysis tools.
Characteristics of a Single-Mode Optical Fibre and Cable	ITU-T Recommendation G.652	International Telecommunication Union (ITU)	Defines standard characteristics of single-mode optical fibers including attenuation, dispersion, and transmission specifications.
Optical Wireless Communication Systems	S. Arnon	Optical Engineering	Explains principles, applications, and challenges of optical wireless communication systems with focus on FSO performance analysis.

Table 2.1 : Literature Review Summary Table

Chapter 3

Methodology

3.1 Introduction

In this section, the methodology adopted for the design and analysis of Free Space Optics (FSO) and Optical Fiber communication systems is described. The methodology includes system design using OptiSystem software, selection of required components, simulation of both communication systems, and analysis of performance under different conditions such as transmission distance and attenuation.

3.2 Design Process

The design process began with outlining the overall architecture of the Free Space Optics (FSO) and Optical Fiber communication systems, focusing on achieving accurate performance analysis and comparison. The initial system design was developed using OptiSystem software, where key components such as PRBS generator, NRZ pulse generator, Mach-Zehnder modulator, optical channel, and receiver were integrated. This simulation phase was important to verify system performance and ensure reliable results before proceeding to detailed analysis.

Firstly, the system parameters were defined, including a data rate of 10 Gbps and a wavelength of 193.1 THz. Both FSO and Optical Fiber systems were designed using similar transmitter and receiver structures to maintain consistency in comparison. The transmission channel was then configured separately for FSO and Optical Fiber, where parameters such as transmission distance and attenuation were varied. For FSO, higher attenuation values were considered to simulate atmospheric conditions, while for Optical Fiber, standard attenuation values were used.

Secondly, performance evaluation mechanisms were incorporated to analyze system behavior. Parameters such as Q-factor, Bit Error Rate (BER), and eye diagram were used to measure signal quality and system efficiency. The eye diagram analyzer and BER analyzer in OptiSystem helped in identifying signal distortion, noise, and performance degradation. After designing and simulating both systems, results were compared to determine the more reliable communication method. The entire design was validated through simulation to ensure accuracy.

and effectiveness of the proposed system.

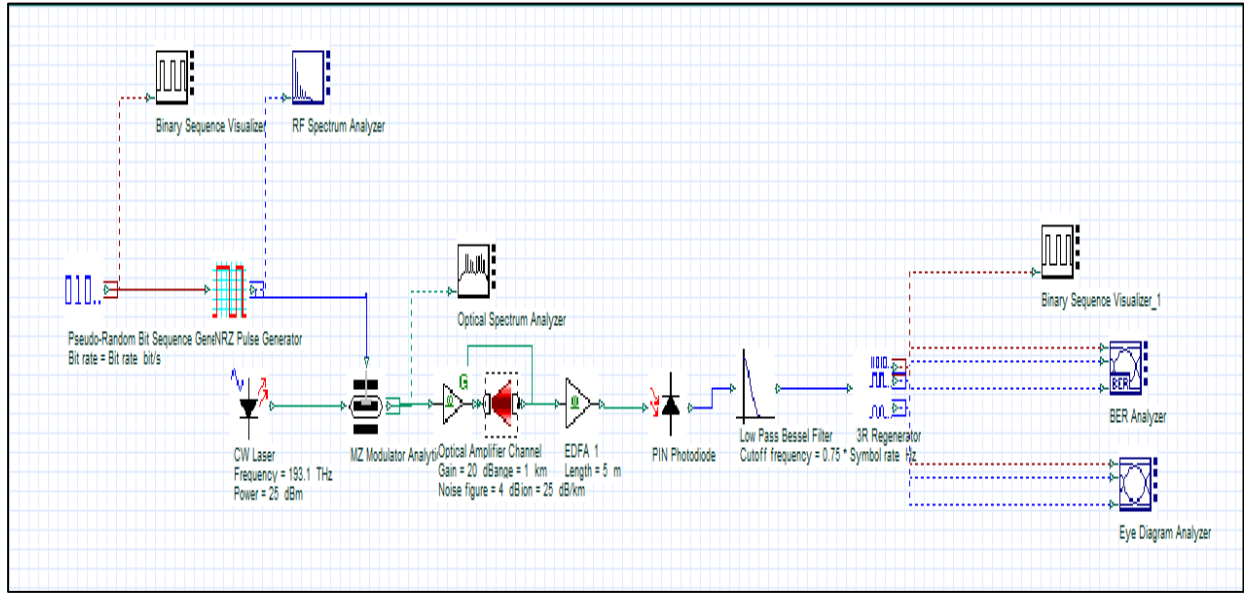


Figure 3.1: Simulation of FSO Channel

As shown in the above figure 3.1, the FSO link is analyzed by varying atmospheric attenuation and transmission distance to evaluate system performance under different environmental conditions. The attenuation is varied as **3 dB/km, 5 dB/km, 10 dB/km, and 15 dB/km**, while the transmission distance is varied as **1 km, 3 km, 5 km, 8 km, and 10 km**.

For each combination of attenuation and range, simulations are performed and the corresponding **Q-factor, BER, and eye diagram** are obtained. This analysis focuses on evaluating the impact of atmospheric losses and link distance on signal quality.

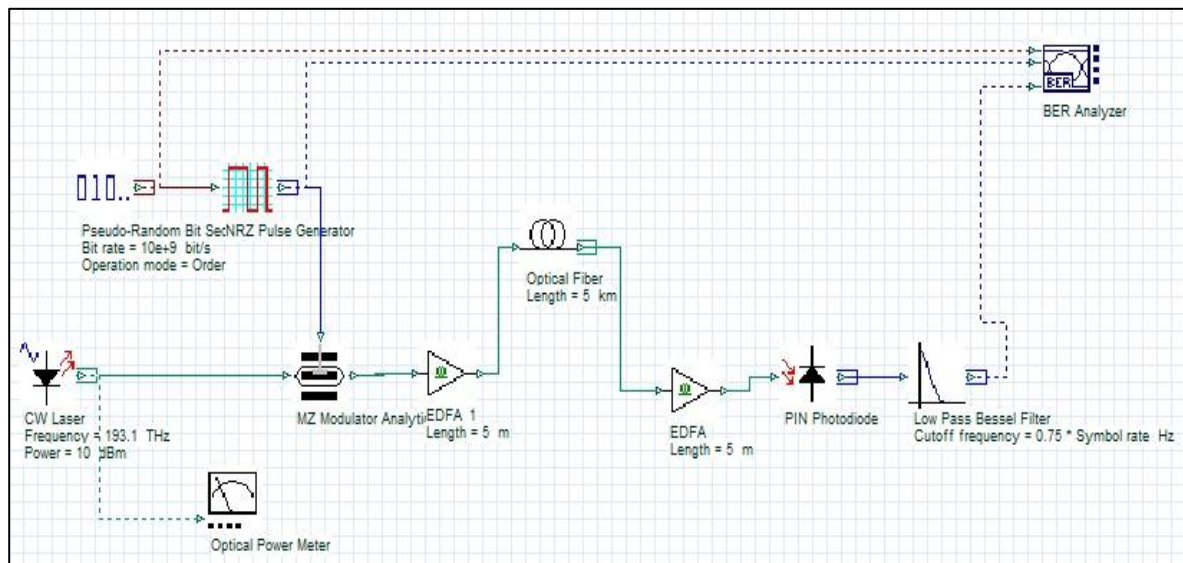


Figure 3.2: Simulation of Optical Fiber Channel

As shown in the above figure 3.2, the optical fiber link, the system is analyzed under a constant attenuation of **0.2 dB/km**, representing standard single-mode fiber conditions. The transmission distance is varied as **2 km, 5 km, 10 km, 20 km, 30 km, and 35 km** to study the effect of link length on system performance.

For each fiber length, simulations are performed and the corresponding **Q-factor, BER, and eye diagram** are obtained. This analysis focuses on evaluating signal degradation with increasing transmission distance under fixed attenuation conditions.

3.3 Algorithm

1. Start
2. Initialize system parameters (laser power, wavelength, bit rate, link distance, and attenuation).
3. Generate pseudo-random binary data using the PRBS generator.
4. Encode the data using an NRZ pulse generator.
5. Modulate the optical carrier using the Mach–Zehnder (MZ) modulator.
6. Transmit the modulated optical signal through the communication channel (FSO or Optical Fiber).
7. Apply channel impairments:

For FSO: atmospheric attenuation and turbulence using Beer–Lambert Law

For Optical Fiber: fiber attenuation and dispersion

8. Receive the optical signal using a PIN photodiode.
9. Pass the detected signal through a Low Pass Bessel Filter.
10. Regenerate the signal using a 3R (Reamplify, Reshape, Retiming) circuit.
11. Measure the Bit Error Rate (BER) using a BER analyzer.
12. Calculate the Q-factor of the received signal.
13. Compare transmitted and received signals to evaluate system performance.
14. Plot the Eye Diagram, BER curves, and Q-factor graphs for analysis.
15. Vary link distance and attenuation, then repeat steps 6–14.
16. Stop

3.4 Performance Metrics

Q-Factor

The **Q-factor (Quality Factor)** is an important parameter used to measure the quality and reliability of a communication system. It indicates how clearly the received signal can be distinguished from noise. A higher Q-factor means better signal quality, lower noise interference, and more accurate data transmission.

It is mathematically expressed as:

$$Q = \frac{I_1 - I_0}{\sigma_1 - \sigma_0}$$

where I_1 and I_0 are the mean signal levels for binary '1' and '0', and σ_1 and σ_0 represent the noise levels. In general, a higher Q-factor results in a lower Bit Error Rate (BER), indicating better system performance. In this project, Q-factor is used to compare the performance of FSO and Optical Fiber systems under different operating conditions.

Bit Error Rate (BER)

The **Bit Error Rate (BER)** measures the accuracy of a communication system by calculating the ratio of incorrect bits received to the total transmitted bits.

It is given by:

$$BER = \frac{\text{Number of Error Bits}}{\text{Total Number of Transmitted Bits}}$$

A lower BER indicates better system performance and more reliable data transmission. BER increases due to factors like noise, attenuation, and distance. In this project, BER is used to compare the performance of FSO and Optical Fiber systems under different conditions.

Eye Diagram

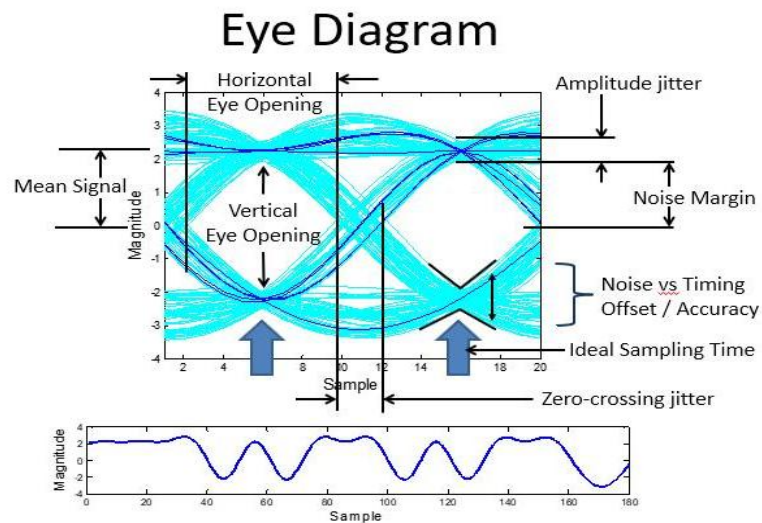


Figure 3.3 : EYE diagram

An eye diagram is a graphical representation of a digital signal in which multiple cycles of a waveform are superimposed on a single time axis using an oscilloscope. This overlapping creates a pattern resembling the shape of an eye, which is why it is called an eye diagram. It is widely used in optical fiber and free-space optical (FSO) communication systems to evaluate signal quality. A well-defined and open eye pattern indicates low noise, minimal distortion, and proper timing synchronization, whereas a closed or distorted eye suggests the presence of noise, intersymbol interference, and timing errors. Key features such as the eye opening, jitter, and noise levels help engineers assess system performance and reliability.

- ☐ **Vertical eye opening** : Indicates noise margin (higher is better)
- ☐ **Horizontal eye opening** : Indicates timing margin
- ☐ **Jitter** : Causes timing variation (horizontal spreading)
- ☐ **Noise** : Causes amplitude variation (vertical spreading)
- ☐ **Ideal sampling point** : Center of the eye for minimum error
- ☐ **Closed eye** : Poor signal quality

In this report, we will be analysing the parameters of FSO and optical fiber communication systems using the eye diagram to study their performance characteristics.

Input Power and Received Power

The **input power** represents the optical power transmitted from the source into the communication system, while the **received power** is the optical power detected at the receiver after transmission through the channel. These parameters are crucial for evaluating the performance of both Free Space Optics (FSO) and Optical Fiber communication systems.

The input power is determined by the laser source and is given by:

$$P_t = \text{Laser Power (dBm)}$$

The received power is affected by channel losses such as attenuation and is calculated as:

$$P_r = P_t - (\text{Attenuation} \times \text{Distance})$$

A higher received power indicates better signal strength at the receiver, leading to improved system performance. As the transmission distance or attenuation increases, the received power decreases due to signal loss in the medium.

Chapter 4 Results and Discussions

4.1 FSO

The results obtained from the simulation were analyzed to evaluate the performance of the communication systems under different conditions. Key parameters such as Q-factor, Bit Error Rate (BER), and received power were observed for varying transmission distances and attenuation values. The outcomes are presented using graphs, tables, and eye diagrams to provide a clear comparison between the systems. The analysis helps in understanding the efficiency, reliability, and limitations of each communication method.

FSO Link

For the simulation of the Free Space Optics (FSO) link, the system was configured as follow:

Bit rate = 10 Gbit/s

CW frequency = 193.1 THz

CW Laser Power = 25 dBm

Symbol rate = 10 G Symbol/sec

Extinction ratio of MZ Modulator = 30 dB

EDFA length = 5 m

The FSO link performance at **3 dB/km attenuation** is shown below:

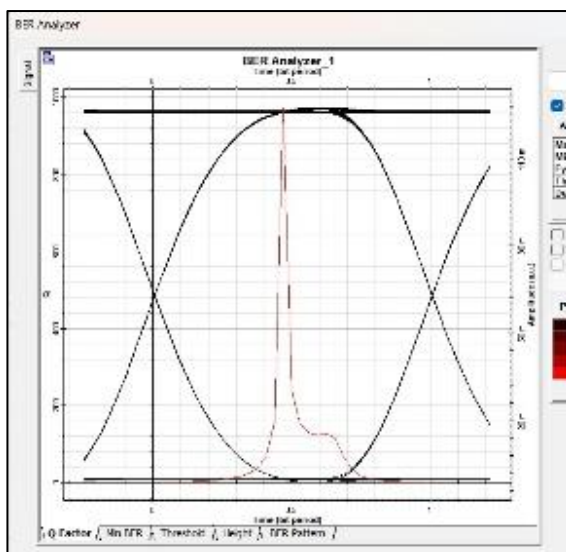


Figure 4.1 Eye Diagram at 1km

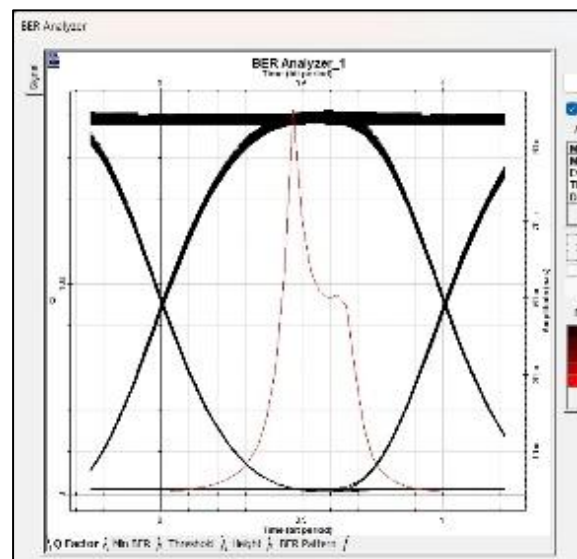


Figure 4.2 Eye Diagram at 3km

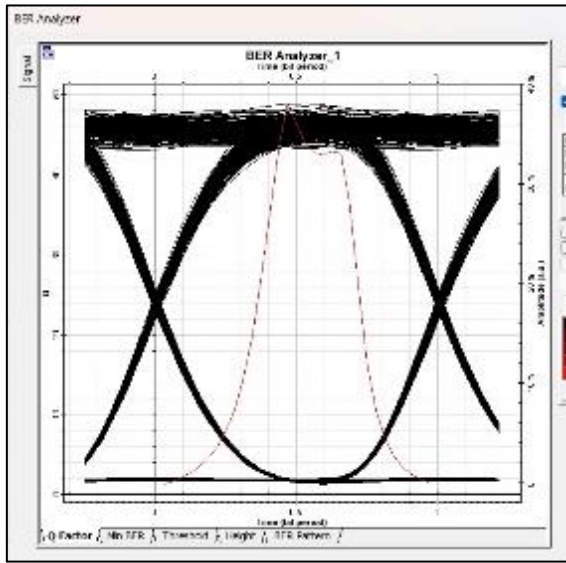


Figure 4.3 Eye Diagram at 5km

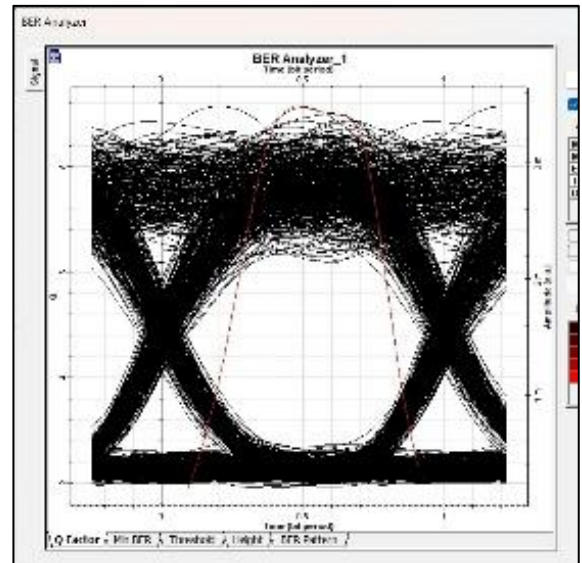


Figure 4.4 Eye Diagram at 8km

Attenuation	Range (Km)	Max. Q factor	Min. BER	Eye Height	Received Power
3 dB/Km	1	553.286	0	0.11758 m	22
	3	90.97	0	0.0642	16
	5	23.53	6.64142e-123	0.01085	10
	8	3.642	0.000121	0.00011	1

Table 4.1 : Performance Metrics Table for Attenuation 3dB/Km

From the results, it is observed that as the transmission range increases from 1 km to 8 km at an attenuation of 3 dB/km, the performance of the Free Space Optics communication system degrades significantly. The Q-factor decreases from 553.286 to 0, the BER increases from 0 to 1, the eye height reduces drastically, and the received power drops from 22 dBm to -5 dBm. This indicates that increasing distance causes higher signal attenuation and weaker signal quality. The system performs efficiently up to around 5 km, while beyond 8 km the communication quality becomes poor and unreliable.

The FSO link performance at **5 dB/km attenuation** is shown below:

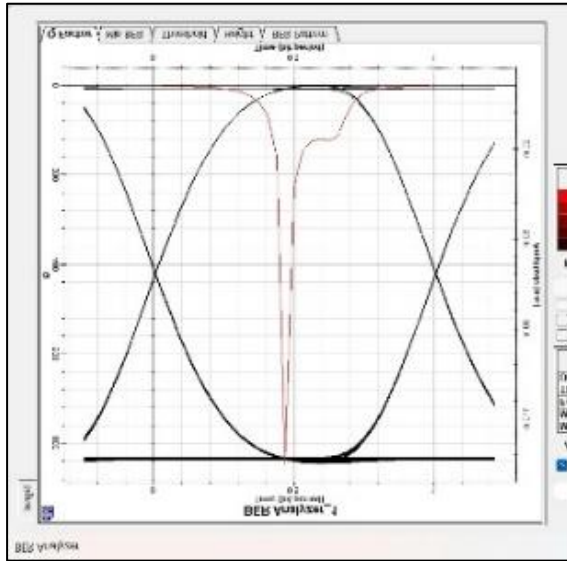


Figure 4.5 Eye Diagram at 1km

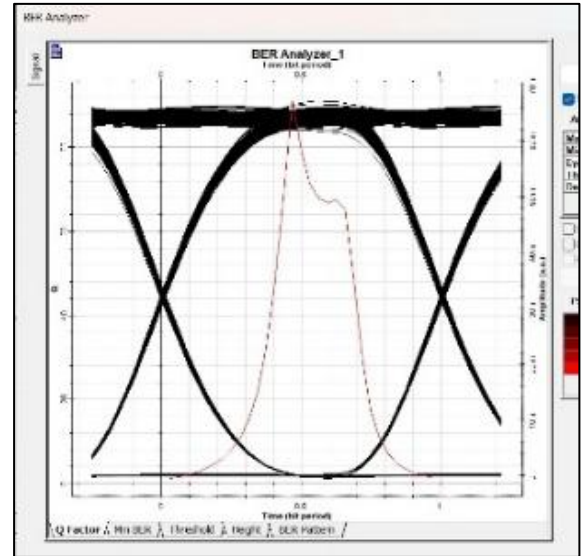


Figure 4.6 Eye Diagram at 3km

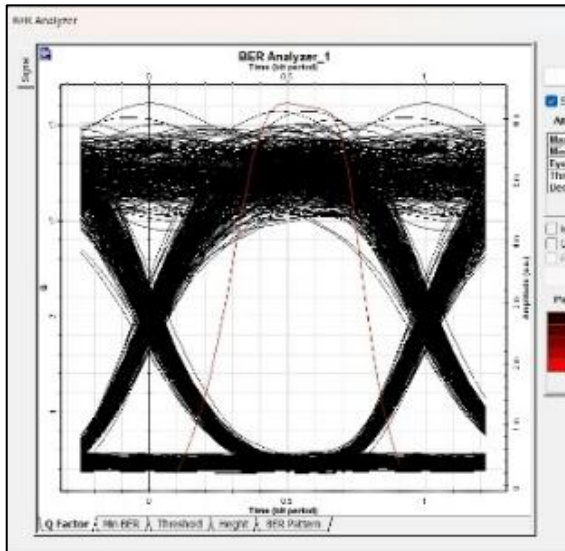


Figure 4.7 Eye Diagram at 5km

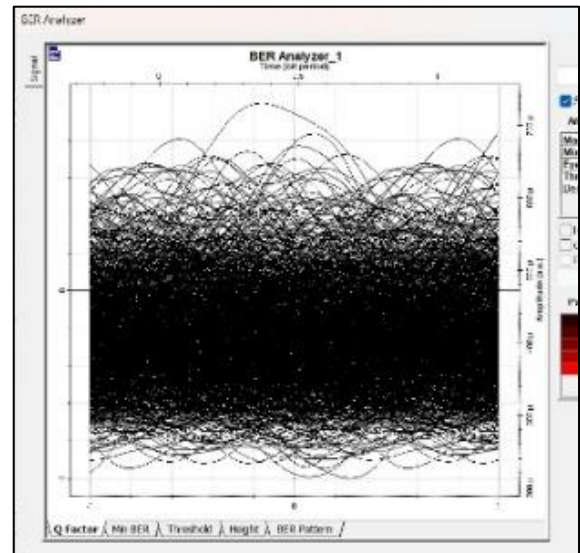


Figure 4.8 Eye Diagram at 8km

Attenuation	Range (Km)	Max. Q factor	Min. BER	Eye Height	Received Power
5 dB/Km	1	467.44	0	0.1153	20
	3	44.11	0	0.02715	10
	5	6.335	1.0266e-10	0.0007	0
	8	0	1	0	-15

Table 4.2 : Performance Metrics Table for Attenuation 5dB/Km

From the results, it is observed that at an attenuation of 5 dB/km, the performance of the Free Space Optics system degrades very rapidly with increase in transmission distance. The Q-factor decreases from 246.366 at 1 km to 0 at 4 km, while the BER rises from 0 to 1, indicating complete communication failure at longer ranges. The eye height and received power also reduce significantly, reaching 0 and -15 dBm respectively at 4 km. The system performs satisfactorily only for short distances up to about 2–3 km under high attenuation conditions, beyond which the signal quality becomes highly unreliable.

The FSO link performance at **10 dB/km attenuation** is shown below:

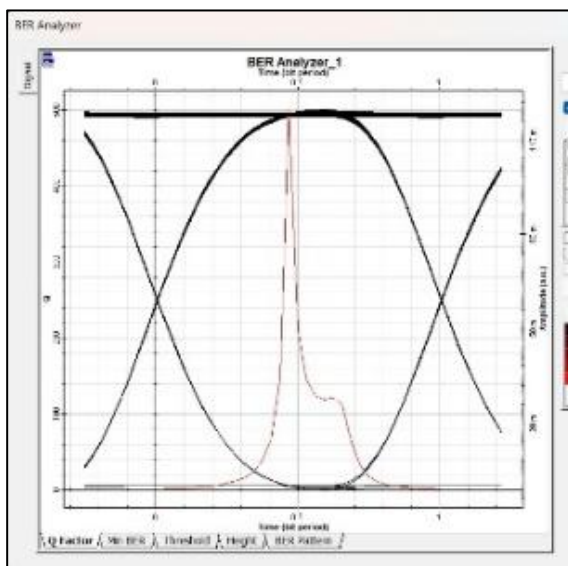


Figure 4.9 Eye Diagram at 1km

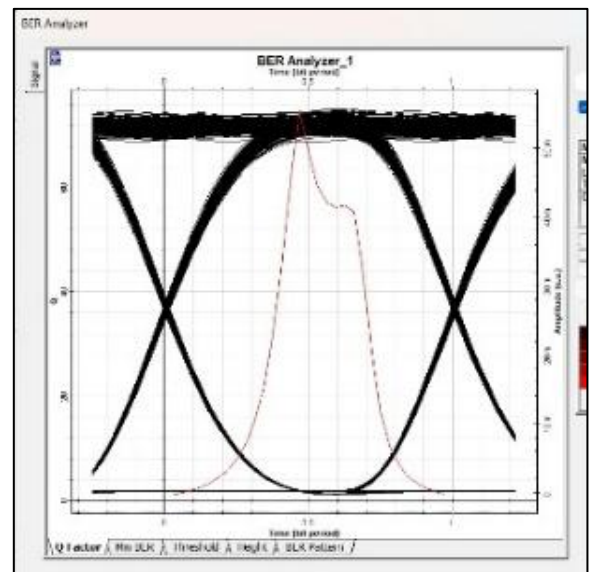


Figure 4.10 Eye Diagram at 2km

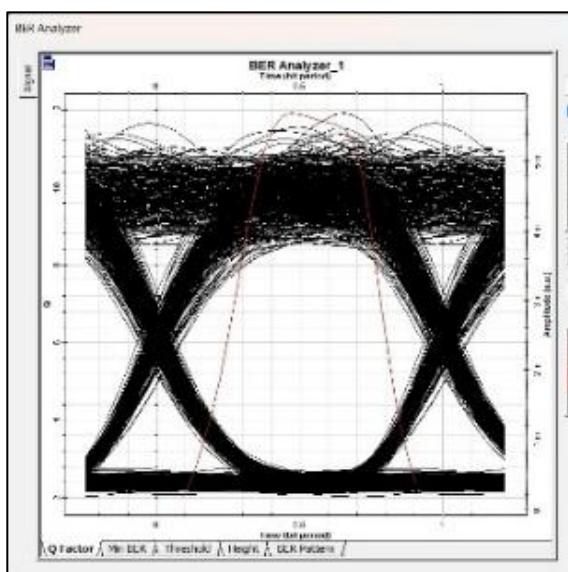


Figure 4.11 Eye Diagram at 3km

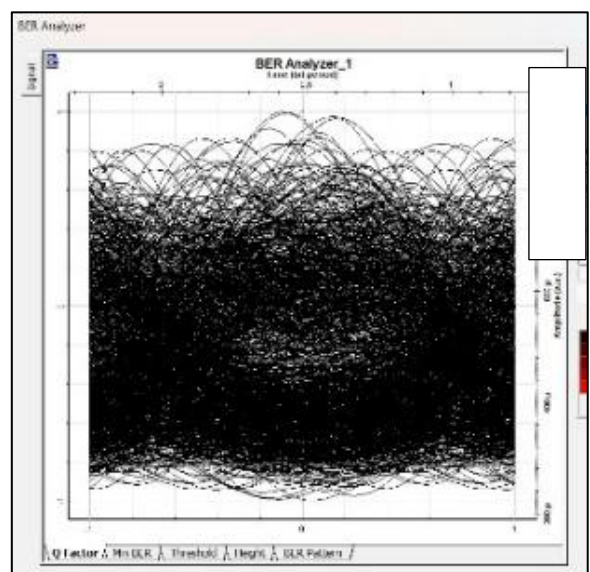


Figure 4.12 Eye Diagram at 4km

Attenuation	Range (Km)	Max. Q factor	Min. BER	Eye Height	Received Power
10 dB/Km	1	246.366	0	0.1054	15
	2	36.4742	9.04024e-292	0.02030	5
	3	5.467	1.84156e-08	0.00054	-5
	4	0	1	0	-15

Table 4.3: Performance Metrics Table for Attenuation 10dB/Km

From the results, it is observed that at an attenuation of 10 dB/km, the performance of the Free Space Optics system degrades very rapidly with increase in transmission distance. The Q-factor decreases from 246.366 at 1 km to 0 at 4 km, while the BER rises from 0 to 1, indicating complete communication failure at longer ranges. The eye height and received power also reduce significantly, reaching 0 and -15 dBm respectively at 4 km. The system performs satisfactorily only for short distances up to about 2–3 km under high attenuation conditions, beyond which the signal quality becomes highly unreliable.

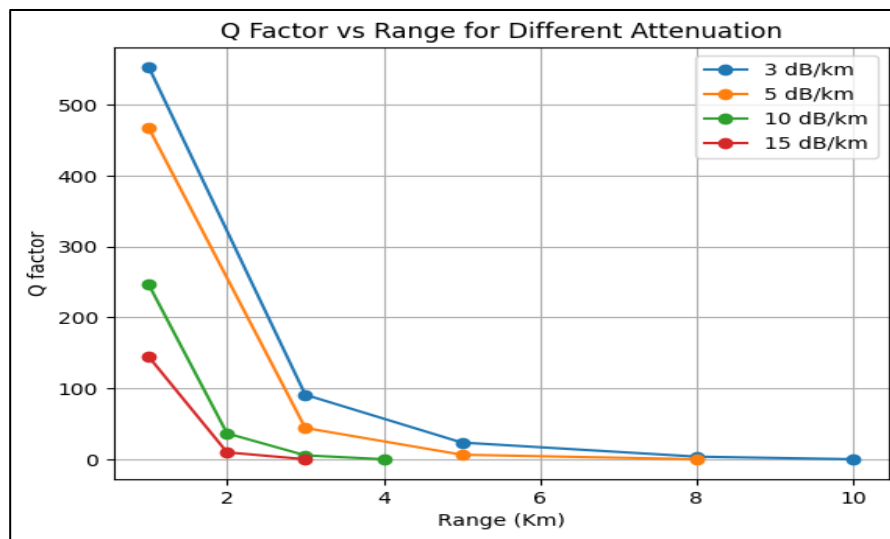


Figure 4.13: Graph of Q-Factor vs Range for FSO

Graph Explanation:

The graph represents the variation of **Q-factor with transmission range** for the **Free Space Optics (FSO) system** under different attenuation conditions. It is observed that as the **range increases**, the **Q-factor decreases significantly** due to atmospheric losses.

For higher attenuation values, the Q-factor drops rapidly and reaches zero at shorter distances, indicating poor system performance. In contrast, lower attenuation conditions maintain a higher

Q-factor over relatively longer distances.

This behavior shows that in FSO systems, **atmospheric attenuation has a major impact on performance**, limiting the effective transmission range.

4.2 Optical Fiber Link

For the simulation of the Optical Fiber link, the system was configured as follow:

Bit rate = 10 Gbit/s

CW frequency = 193.1 THz

CW Laser Power = 25 dBm

Symbol rate = 10 G Symbol/sec

Extinction ratio of MZ Modulator = 30 dB

EDFA length = 5 m

The Optical Fiber link performance at **0.2 dB/km attenuation** is shown below:

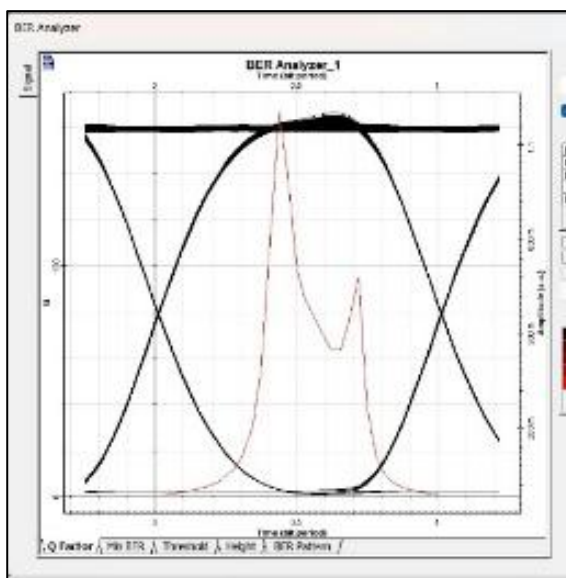


Figure 4.14 Eye Diagram at 2km

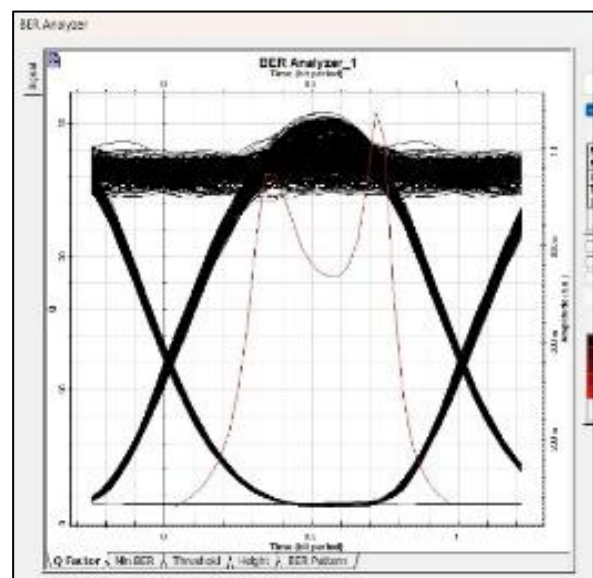


Figure 4.15 Eye Diagram at 5km

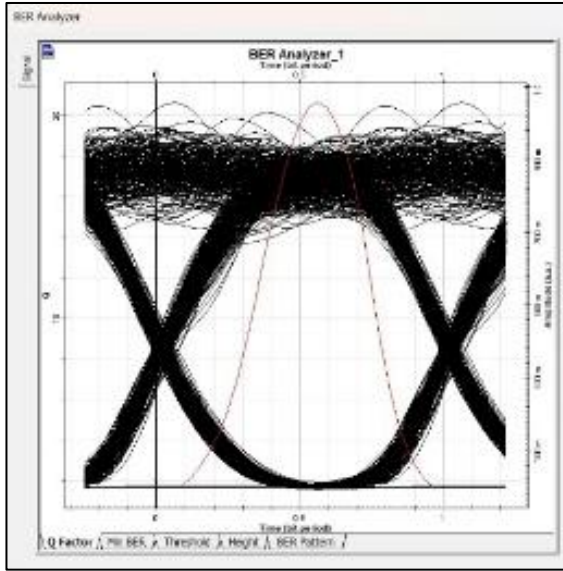


Figure 4.16 Eye Diagram at 10km

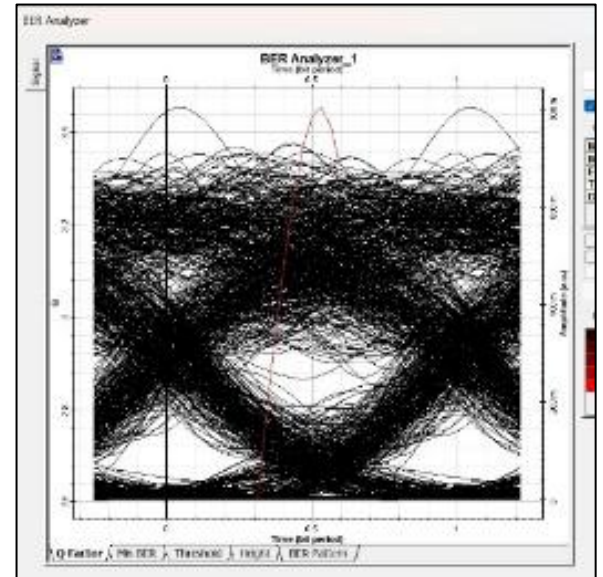


Figure 4.17 Eye Diagram at 20km

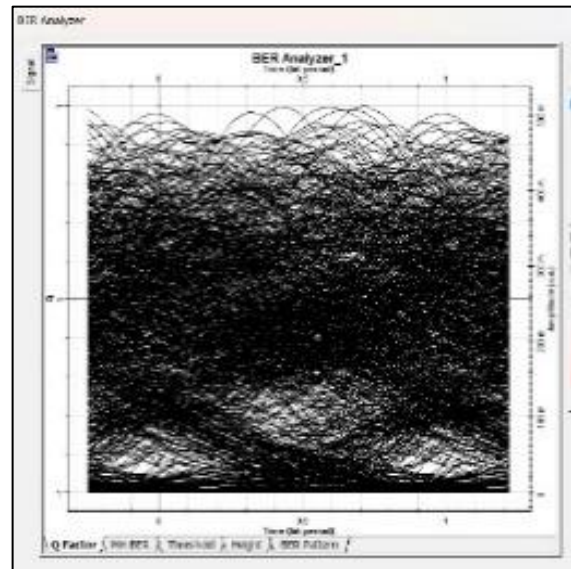


Figure 4.18 Eye Diagram at 30km

Attenuation	Range (Km)	Max. Q factor	Min. BER	Eye Height	Received Power
0.2Db/Km	2	271.43	0	0.40607	24.6
	5	76.918	0	0.3653	24
	10	37.5327	1.15889e-308	0.30044	23
	20	17.37	4.28374e-68	0.21808	21
	30	5.63642	7.25109e-09	0.0935	19

Table 4.4: Performance Metrics Table for Attenuation 0.2dB/Km

From the results, it is observed that in Optical Fiber Communication (OFC) with an attenuation of 0.2 dB/km, the system maintains good performance even at longer transmission distances.

Although the Q-factor gradually decreases from 271.43 at 2 km to 5.636 at 30 km, the BER remains extremely low, indicating reliable communication. The eye height and received power also decrease slowly with increasing distance, but the received power remains comparatively high even at 30 km. This shows that OFC systems experience very low attenuation losses and can support long-distance communication with stable signal quality compared to FSO systems.

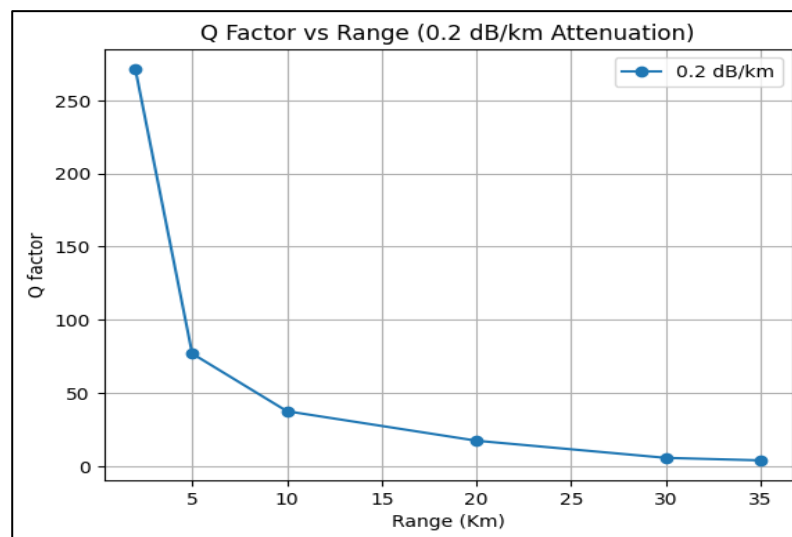


Figure 4.19: Graph of Q-Factor vs Range for Optical Fiber

Graph Explanation:

The graph represents the variation of **Q-factor with transmission range** for the **Optical Fiber communication system** at an attenuation of **0.2 dB/km**. It is observed that as the **range increases**, the **Q-factor gradually decreases** due to fiber attenuation.

However, the decrease is relatively slow compared to FSO systems, indicating better signal preservation over long distances. Even at higher ranges, the Q-factor remains at acceptable levels, showing stable system performance.

This demonstrates that optical fiber communication provides **reliable transmission with low attenuation**, making it suitable for long-distance communication.

4.3 Summary of Results:

FSO (Free Space Optics) Summary

In Free Space Optics, performance decreases rapidly with increase in distance and attenuation. The Q-factor drops sharply, BER increases, eye height reduces, and received power becomes

very low. Hence, FSO is suitable only for short-range communication and is highly affected by atmospheric losses.

OFC (Optical Fiber Communication) Summary

In Optical Fiber Communication, performance remains stable even over long distances. The Q-factor decreases slowly, BER stays very low, and received power remains high. Hence, OFC is reliable for long-distance communication with minimal signal degradation.

Observation:

- Q-factor decreases as distance and attenuation increase due to signal loss.
- In FSO systems, performance degrades rapidly at higher attenuation values, and Q-factor becomes zero at shorter distances.
- Optical fiber maintains stable performance even at longer distances due to low attenuation (0.2 dB/km).
- Received power decreases with increase in distance, leading to reduced signal quality.
- FSO shows significant power loss due to atmospheric effects, whereas optical fiber shows minimal loss.

(a) Q-factor vs Distance

The graph of Q-factor versus transmission distance shows a decreasing trend. In FSO systems, the Q-factor drops sharply with distance due to atmospheric attenuation, whereas in optical fiber, the decrease is gradual, indicating better performance over long distances.

(b) Q-factor vs Attenuation

As attenuation increases, the Q-factor decreases significantly. For higher attenuation values (e.g., 10–15 dB/km in FSO), the Q-factor quickly approaches zero, indicating poor system performance. Optical fiber shows stable Q-factor due to very low attenuation.

(c) Received Power vs Distance

Received power decreases with increasing distance due to attenuation in the transmission medium. It is given by:

$$P_r = P_t - (\alpha \times L)$$

where,

P_r = Received Power

P_t = Transmitted Power

α = Attenuation (dB/km)

L = Distance (km)

In FSO systems, the received power decreases rapidly due to high atmospheric losses, while in optical fiber, the decrease is slow, ensuring reliable communication over longer distances.

Chapter 5

Conclusion & Future Scope

5.1 Conclusion

In this paper, the performance of Free Space Optics (FSO) and optical Fiber communication systems has been analyzed and compared using simulation-based modeling. The study clearly shows that both technologies have their own advantages as well as limitations depending on the operating conditions.

The results indicate that FSO communication provides high data rates and flexibility in deployment, making it suitable for short-range and quick-installation scenarios. However, its performance is highly affected by atmospheric conditions such as fog, rain, and turbulence, which lead to signal degradation, reduced Q-factor, and increased Bit Error Rate (BER) as the transmission distance increases.

In contrast, optical Fiber communication demonstrates consistently superior performance with high Q-factor, negligible BER, and stable signal transmission even over longer distances. The guided nature of Fiber ensures minimal external interference and lower attenuation, making it a highly reliable solution for long-distance and high-capacity communication systems.

Overall, the comparative analysis highlights that while Fiber optic communication is more reliable and efficient for stable, long-term networks, FSO remains a practical alternative in situations where Fiber deployment is difficult or not feasible. Therefore, the choice between FSO and Fiber depends on application requirements, environmental conditions, and infrastructure constraints.

This study provides a clear understanding of the performance trade-offs between the two technologies and can serve as a useful reference for selecting appropriate communication systems in real-world applications.

5.2 Future Scope

The present work focuses on the performance analysis of FSO and optical fiber communication systems under varying attenuation and distance conditions. However, several improvements and extensions can be considered for future research:

- Implementation of a **hybrid FSO–Optical Fiber system** to combine the advantages of both technologies for improved reliability.
- Analysis under **real atmospheric conditions** such as fog, rain, and turbulence to obtain more practical results for FSO systems.
- Use of advanced modulation techniques (such as **QAM or DPSK**) to enhance data rate and system performance.
- Integration of **error correction techniques (FEC)** to reduce BER and improve signal quality.
- Performance evaluation using **different wavelengths and power levels** for optimized system design.
- Implementation of **adaptive systems** that can switch between FSO and fiber links based on channel conditions.
- Investigation of **long-distance communication with optical amplifiers (EDFA)** for improved signal strength.

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